

MultiPlanetary Exchange

Evidence Appendix - Ceres Clearing House

Evidence basis. This appendix uses only the project brief and the team-generated Ceres-hub evidence available in the submission workspace. Conditional traces are not unconditional delivery promises. Missing engine exports are identified rather than invented.

Coverage map

Item	What this appendix contains	Status
S1	17-scenario test summary; worked cross-planet, retry, futures and margin evidence; scale/binding sensitivity	COMPLETE: final S1 engine export reports per-run completion, peak encumbrance/utilization, cash-hours, share-hours, settled value, quota packets, and launches.
S2	72 h clearing outage result; recovery duration; margin sensitivities	SUPPORTED
S3	Nine-settlement service rule + best/median/worst logic	COMPLETE: final S3 exports include Ceres h=0 and h=300 routes for all nine settlements plus best/median/worst full timelines.
E1	Orbital method + epoch validation errors	SUPPORTED except numeric moving-receiver worked example
E2	Launch-loss/retry probabilities; traffic totals; Ceres-vs-Earth packet cost	SUPPORTED
E3	Opening balances, margin/capital sensitivity, conservation/invariants	SUPPORTED; per-step ledger and obligations exported for all 18 runs
E4	200-year Ceres/Earth hub robustness metrics	SUPPORTED; 5,748 refined event boundaries exported
E5	Ceres batch lengths at +1/+10/+100 y	SUPPORTED; route table at shifted epochs exported

S1. Scenario evidence

The Ceres-hub engine reports 17/17 scenarios passing, 10,146 invariant checks, cash and shares conserved in every run, six attempted double spends blocked, zero duplicate settlements, and six repeat deliveries safely ignored.

Scenario	Checks	Evidence point
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Local trade	9/9	Local continuous execution
Cross-planet trade	11/11	Batch + \$47.50 midpoint + direct settlement
Equal-price auction	6/6	Source stamp beats arrival time
Exact tie / pro-rata	4/4	Deterministic exact-tie allocation
Partial fill	5/5	Unused reservation released
Late order	9/9	Rollover without changing limit
Limit protection	8/8	No fill outside stated limit
Cancel before / after	6/6; 4/4	Cancel only before execution
Retry / duplicate	8/8	Three losses; fourth launch lands; executes once
Uncertain locking	6/6	Backing stays locked while outcome uncertain
Settlement finality	4/4	Destination finality usable
Futures opening	7/7	Funded opening
Margin call	8/8	Location-aware deadline
Funded default	8/8	No unbacked loss
Disconnection	5/5	Local-safe activity continues
Clearing outage / recovery	5/5	Ceres outage handled and service restored

S1-A. Cross-planet value move

Measure	Ceres-hub result
Batch length at hour 0	30.68 h
Cross-planet trade complete	38.5 h from order to both settlement legs usable
Clearing example	\$45 sell vs \$50 buy clears at midpoint \$47.50
Neptune fairness	Neptune can arrive ~4.1 h after Ceres-local bidder and still wins an equal-price tie if its trusted source stamp is earlier
Settlement	Direct operator-to-operator after match; Earth->Mars 0.6 h vs 1.1 h if relayed through Ceres

Binding transport variation: the Neptune slowest hop loses 42.7% of individual launches at the tested geometry. With 0/1/2/3 retries covered, batch length is 4.5/13.2/22.0/30.7 h; only the three-retry setting admits the transport's full four-launch sequence in the same batch.

S1-B/C. Price-dependent futures runs

Rule / run	Evidence
Contract scale used in margin experiment	10 MOI futures × entry 100 × multiplier 100 = \$100,000 notional
Initial margin rule	5% per 12 h observation, adjusted for

	communication-risk window
Neptune long IM	\$15,000
Earth short IM	\$10,000
Falling path	5 calls; lowest cushion \$8,300; 50 quota packets; peak cash locked \$93,700
Crash / nonpayment path	Guarantee fund used \$7,000; unbacked amount \$0
Scale / binding variation	3% margin lowers peak locked to \$85,020 but creates 7 calls; 8% raises peak to \$106,720 and reduces calls to 3

S1 resource metrics. S1 is now backed by the final engine export: 18 runs (17 scenarios plus the rising-price future) with completion time, peak encumbered cash, utilization, cash-hours, share-hours, settled value, quota packets and launches. Margin call peaks at \$93,700 (30.2%); the rising-price future peaks at \$97,000 (31.3%).

S2. Stress and recovery

The team chose a 72-hour clearing-house outage because it attacks the architecture's most centralized institution: Ceres Clearing. With Ceres as hub, the measured period of lost financial service is 80.7 h, including recovery to resumed financial service rather than merely first packet receipt.

During outage	Financial state
Existing funded positions	Remain recorded and collateralized
New futures risk	Paused
Local same-settlement activity	May continue when it does not require Ceres
Uncertain remote settlement	Backing remains locked
Recovery	Restore sessions -> exchange state -> deduplicate -> reconcile -> process overdue margin/maturity -> reopen global services

Sensitivity under identical funding

Variation	Calls	Quota packets	Crash fund used	Interpretation
Maintenance 50%	2	41	\$10,000	Few calls; larger crash draw
Maintenance 75% - chosen	5	50	\$7,000	Middle quota/risk tradeoff
Maintenance 100%	16	83	\$0	Strongest crash cushion; highest traffic

The available Ceres evidence does not include a separate numeric scheduled-maintenance-removed rerun for this exact outage. That specific S2 comparison remains an engine-export item.

S3. Access at every settlement

Every settlement receives local continuous equity service. Cross-planet equity orders reach the Ceres batch through the best usable Relay A/B route; futures require Ceres Clearing; the MOI price source originates at Mars. The global batch is sized by the slowest eligible participant, so Ceres-local users do not receive arrival-time priority.

Settlement	Local equity	Cross-planet / futures access	Known Ceres margin window where measured
Mercury	Continuous	Relay route to Ceres	Not measured
Venus	Continuous	Relay route to Ceres	Not measured
Earth	Continuous	Relay route to Ceres	2.74 h
Mars	Continuous	Relay route to Ceres; MOI source local	2.49 h
Ceres	Continuous	Batch/clearing local	0.00 h
Jupiter	Continuous	Relay route to Ceres	Not measured
Saturn	Continuous	Relay route to Ceres	Not measured
Uranus	Continuous	Relay route to Ceres	Not measured
Neptune	Continuous	Relay route to Ceres; worst tested latency/loss	16.93 h

S3 route export. The brief requires exact best route, one-way delay, and 24-hour availability at h=0 and h=300 for all nine settlements, plus best/median/worst full timelines. Final Ceres route exports are now available at h=0 and h=300 for all nine settlements. All best routes shown in those snapshots have 100% 24-hour route availability; Neptune is the slowest client at about 4.11-4.12 h one way.

E1. Orbital and light-time calculations

Team orbital evidence uses Sun-centered geometric 3D J2000 ecliptic coordinates in AU and continuous 86,400-second days. Propagation advances mean anomaly, solves Kepler's equation by Newton iteration, converts orbital-plane coordinates, then rotates by argument of perihelion, inclination, and ascending node. Light time is solved against the receiver's position at arrival by fixed-point iteration; forward and return messages are solved separately.

Body	Maximum epoch-position error (AU)
Mercury	1.57e-7
Venus	2.15e-7
Earth	3.84e-7
Mars	4.68e-7
Ceres	3.89e-7
Jupiter	3.98e-7
Saturn	3.39e-7

Uranus	4.51e-7
Neptune	2.75e-7
Relay A / B	4.44e-16

The supplied epoch-validation file supports the epoch check. A separate numeric worked example isolating the moving-receiver correction was not present in the supplied Ceres evidence and is therefore not invented here.

E2. Probability and traffic

Origin / slowest hop to Ceres	Launch lost	Need >1 retry	Need >2	All 4 launches fail
Earth -> Relay A	4.4%	0.2%	~0.0%	~0.0%
Mars -> Relay A	3.6%	0.1%	~0.0%	~0.0%
Neptune -> Relay A	42.7%	18.2%	7.8%	3.3%

If a hop exhausts all four launches, the current batch cannot treat the financial instruction as delivered. Its economic state remains reserved/pending; an endpoint retry may carry the same transaction identity to a later batch.

Traffic comparison over same 17 scenarios	Earth hub	Ceres hub
Backbone quota packets	178	234 (+31%)
Backbone launches	1,463	1,966 (+34%)
Shared quota	600 per 24 h	600 per 24 h

The higher Ceres traffic comes from the Earth-heavy balance sheet: Earth margin/fund cash must travel to Ceres and Earth winners receive remote payouts. This is an accepted design cost, not an orbital advantage.

E3. Balances and capital

Account	Location	Opening cash	Opening shares
E-Alice	Earth	\$70,000	600
E-Bob	Earth	\$45,000	400
M-Carla	Mars	\$60,000	550
M-Diego	Mars	\$45,000	450
N-Eve	Neptune	\$55,000	500
C-Finn	Ceres	\$35,000	500
TOTAL		\$310,000	3,000

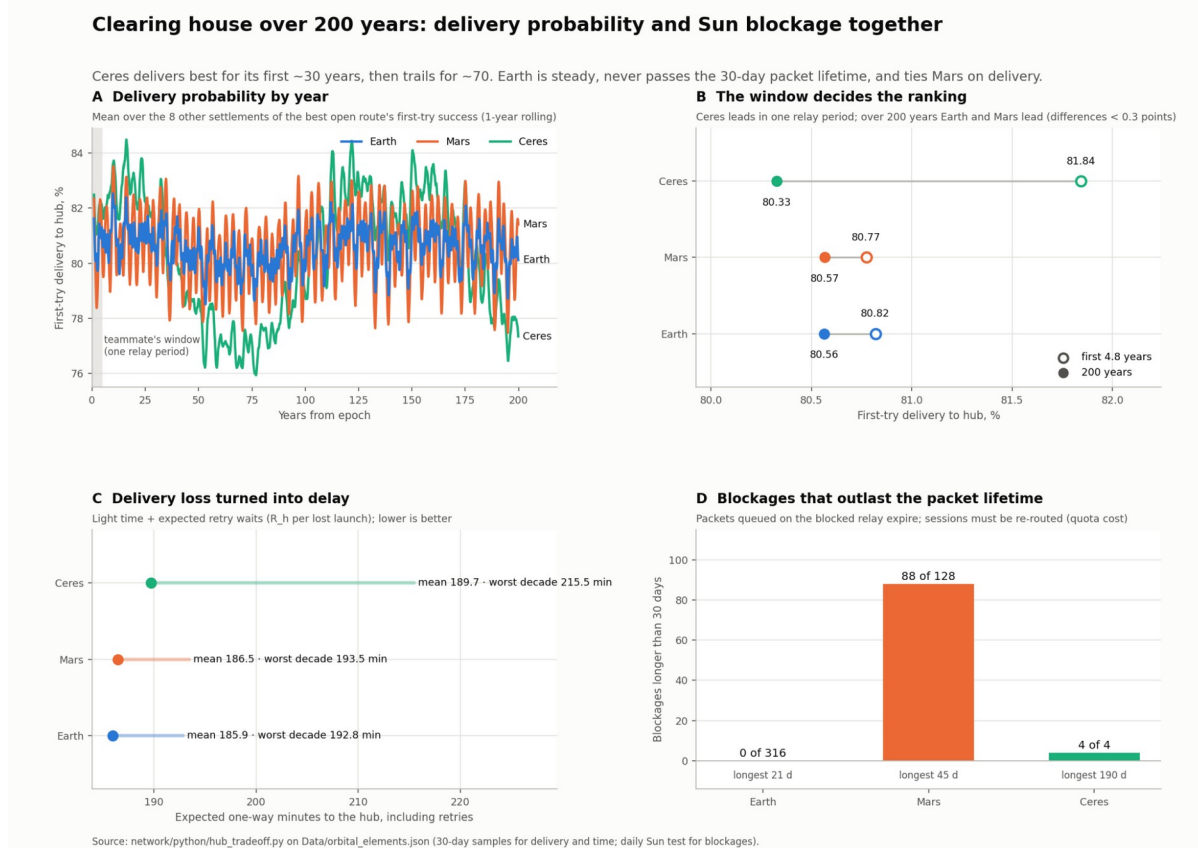
Engine invariants report cash and shares conserved in every scenario, six attempted double spends blocked, zero duplicate settlements, and six repeat deliveries safely ignored. In the falling futures path, peak cash locked is \$93,700 under the chosen 5% margin setting.

E3 ledger export. `traces/<scenario>_ledger.csv` records every balance change in all 18 runs with owner, location, asset, source, encumbrance and liability; `_steps.csv` lists open obligations per step. Cash and shares are conserved at every step in 18/18 runs; no over-encumbrance.

E4. Long-horizon scan - Ceres hub

The team ran a 200-year network comparison and selected Ceres for lower exposure to single-relay operation. These are model-scan results, not all-time mathematical guarantees.

200-year factor	Earth	Ceres
Delivery probability with proactive relay switching	94.23%	94.28%
Time operating with one relay blocked	8.6%	0.8%
Planned relay re-routing events	316	4
Longest observed one-relay blockage	21 days	190 days
Worst-decade expected delay incl. retries	192.8 min	215.5 min
Mean batch length (range)	31.9 h (29.9-35.7)	32.1 h (29.7-36.1)



Long-run Earth/Mars/Ceres hub tradeoff generated by the team network analysis.

Interpretation: Ceres enters a one-relay state much less often, but its rare one-relay blockage spells can be long. The production design therefore calls for proactively moving sessions to the alternate relay before predicted blockage; the current demo does not yet implement that re-pin/resubmission behavior for every shifted-epoch closure. E4 refines 5,748 Sun-closure boundaries to one-second resolution; all 5,748 flip across $t-1$ s / t / $t+1$ s. No settlement loses both relays in the modeled 200 years. Neptune reaches 5.15 h maximum one-way delay; Ceres has four

closures and a 190.2-day longest one-relay spell. E5 demonstrates why automatic route migration is needed for liveness.

E5. Shifted epochs

Offset	Earth-hub batch	Ceres-hub batch	Ceres result
+1 year	34.2 h	34.1 h	Conditional service
+10 years	31.4 h	31.4 h	Conditional service
+100 years	33.0 h	33.3 h	Conditional service

The hub comparison advances the network geometry and reports that Ceres remains within 2% of Earth on the shifted batch metric. E5 now tests +1, +10, +100 years and three difficult epochs. The important difficult-epoch result is a liveness limitation: when Ceres loses a relay, the futures remain funded/owned but open past maturity because proactive session re-pinning and abandoned-message resubmission are not implemented.

Evidence limits and submission note

The 17 scenario traces are conditional except where loss/outage is deliberately forced. Price paths are scripted test inputs. Idle-session expiry and detailed link queues/capacity are not simulated in the team evidence. Relay placement is fixed by the brief.

Final evidence exports received. S1, S3, E4 and E5 are now backed by executable workspace outputs. The difficult E5 liveness limitation is retained openly rather than repaired in the paper.

Sources

1. MultiPlanetary Exchange System - Participant Brief, 2 Oct 2026.
2. Team-generated design-choices evidence, 3 Oct 2026 (Ceres-hub generation).
3. Team-generated Earth-vs-Ceres hub comparison, 3 Oct 2026.
4. Team hub tradeoff / blockage plots and epoch-validation output.

Final engine evidence addendum

The following replaces the earlier evidence-gap notices. Reproduce from repository root with `python3 -m evidence.appendix s1, traces, s3, e4, e5`. The full suite after generation reports 32/32 tests passing.

S1 final resource metrics

Run	Peak cash	Util.	Cash-hours	Share-hours	Quota	Launches
Cross-planet	\$5,000	1.6%	171,452	3,074	8	64
Retry / duplicate	\$5,000	1.6%	171,452	3,074	10	81
Futures opening	\$75,000	24.2%	813,244	0	8	64
Margin call	\$93,700	30.2%	25,989,277	0	50	400
Funded default	\$75,000	24.2%	14,731,741	0	36	288
Clearing outage	\$75,000	24.2%	4,360,660	0	10	149

Rising future	\$97,000	31.3%	27,709,626	0	53	424
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S3 final Ceres routes and timelines

Snapshot	Earth -> Ceres	Mars -> Ceres	Neptune -> Ceres	24 h availability
h=0	0.556 h	0.495 h	4.11 h	100% on selected routes
h=300	0.537 h	0.508 h	4.12 h	100% on selected routes
Client	Batch close	Shares usable	Cash usable / completion	
Ceres best	h 5.99	h 7.10	h 7.10	
Jupiter median	h 6.33	h 7.74	h 7.87	
Neptune worst	h 31.18	h 35.91	h 39.46	

Worst-client trace: Earth and Neptune orders are stamped/reserved at h 1.0003; they reach Ceres at h 1.5565 and h 5.1053. The batch executes at h 31.1844 at \$47.50. Neptune shares become final/usable at h 35.9124 and Earth cash at h 39.4642.

E4 refined 200-year evidence

Settlement	Closures	Shortest	Longest	Avail A/B	Both blocked	Delay min/med/max
Mercury	1577	6.7 d	11.1 d	90.95/90.94%	0 d	0.34/0.72/1.50 h
Earth	316	19.3 d	20.6 d	95.69/95.69%	0 d	0.28/0.72/1.53 h
Mars	128	27.7 d	45.2 d	96.88/96.88%	0 d	0.21/0.74/1.56 h
Uranus	80	22.2 d	23.8 d	98.73/98.73%	0 d	2.24/2.97/3.82 h
Neptune	82	13.2 d	22.0 d	99.03/99.03%	0 d	3.78/4.47/5.15 h

All 5,748 refined Sun-closure boundaries flip at one-second resolution. No settlement loses both relays during the modeled 200 years. Ceres itself has four closures; the longest one-relay spell is 190.2 days.

E5 shifted epochs and implementation finding

Epoch	Batch	Value move	Falling future	Rising future
+1 y	34.08 h	11/11, h 44.04	settled h 300.5	settled h 300.5
+10 y	31.40 h	11/11, h 40.59	settled h 300.3	settled h 300.3
+100 y	33.33 h	10/11, h 43.37	settled h 300.8	settled h 300.8
Hub loses relay	35.59 h	10/11, h 45.90	OPEN past h 2880	OPEN past h 2880
Slowest client	35.47 h	11/11, h 45.89	settled h 301.2	settled h 301.2

Most links closed	32.04 h	10/11, h 41.29	settled h 300.8	settled h 300.8
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The 10/11 value-move rows fail only a check hard-coded to the hour-0 pinned route; trades, invariants and conservation are unaffected. At the hub-loses-relay epoch, 17 of 25 price marks are abandoned after four end-to-end attempts, including maturity. An alternate route exists, but proactive re-pinning/resubmission is not implemented, so the funded position remains open. This is a liveness limitation, not a double-spend or conservation failure.